

Практикум С6

123 marks of the archive, seven techniques, twelve tasks. Everything the archive asks about planes from May 2021 to November 2025: the equation of a plane and its normal, where a line meets a plane, the line where two planes meet, three planes — and the systems of equations they are — and the perpendicular from a point: its foot, its length and the reflection.

О практикуме

Разрез продолжает С5: всё, где плоскость — сам предмет вопроса. Уравнение и нормаль, встреча прямой с плоскостью, прямая пересечения двух плоскостей, три плоскости и системы уравнений, перпендикуляр из точки и отражение. 38 баллов из 182 geometry.vectors (0.209) и 98 из 216 geometry.vectors_3d (0.4537), разобрано по-блочко. 136 баллов по корпусу, 123 рабочих — ноябрь 2023 Q12(b)–(d) стоит дважды. 51 балл Paper 1, 72 Paper 2.

Заданий

0

Раздел решений

включён

Интерактивная версия

kaggle.com/code/artemkonukhov/ib-math-praktikum-c6-planes

Собрано из ноутбука репозитория `ib-math-aa-hl-past-papers`. Проверки в PDF не работают — они для интерактивной версии.

C6 — Planes: where lines and planes meet

123 marks of the archive, seven techniques, twelve tasks. Everything the archive asks about planes from May 2021 to November 2025: the equation of a plane and its normal, where a line meets a plane, the line where two planes meet, three planes — and the systems of equations they are — and the perpendicular from a point: its foot, its length and the reflection.

Paper 2 carries 72 of the marks and Paper 1 carries 51, and on both the work is the same: a normal, a substitution, one linear equation.

The one idea

A plane is **a set of points**, just as a line is, and every question about it is a question about which points belong to it.

$$\mathbf{r} \cdot \mathbf{n} = \mathbf{a} \cdot \mathbf{n}$$

is one point $\mathbf{a}$ of the plane and one direction $\mathbf{n}$ across it. Any other point of the plane and any multiple of the normal describe **the same plane**.

And what follows from it

the question asks	what it is
the equation of a plane	any point on it, any normal — up to a factor
where a line meets a plane	the points of the line that satisfy the plane's equation
the line of intersection of two planes	the points on both
solve a system of three equations	the points on three planes: one, a line of them, or none
the reflection of a point	the point as far beyond the plane as the point is in front of it

How the checks work

They do not know the answers. A plane is handed to them **the way the question builds it**, and a point is handed **the conditions** that fix it:

```

P3 = plane(R, P1, P2)           # through R,
perpendicular to P1 and P2
verify_plane('4', q4, P3)

C = unknown('C', 3)
verify_find('l1c(i)', q11c_i, C, [on(C,
plane_P), perpendicular(C - B, plane_P)])

```

The first is *"is your equation the plane through R perpendicular to both?"* — any multiple of it passes. The second is *"is your C on the plane, with $\overrightarrow{BC}$ along the normal?"*: the check solves the conditions itself.

When you are wrong the check says **how**:

what you wrote	what the check says
$(14, 21, -7)$ for $\overrightarrow{AB} \times \overrightarrow{AC}$	the middle component of a vector product has the wrong sign
the right-hand side $-\mathbf{n} \cdot \mathbf{a}$	the right-hand side has the wrong sign
the normal of Π_1 as the direction of the line where Π_1 and Π_2 meet	the line lies in Π_1 : its direction is across that normal
the value of λ where the point is asked	put it into the line
the foot of the perpendicular for the reflection	the reflection is as far again beyond it
$ \lambda $ for the distance	the distance is $ \lambda \mathbf{n} $

Order of work

level	what it means	tasks
	a plane and its normal	1-4
	a line and a plane, two planes, three planes and systems	5-9
	the perpendicular: its foot, the distance, the reflection	10-12

Every task is a real past-paper question, cited.

72 of the 123 marks are on a calculator paper, and the calculator does almost none of the work. It solves a system with numbers in it — the markscheme allows that — and takes a square root. A system with a letter, a normal, a foot and a reflection are done by hand on either paper.

```

import sys
sys.path.append('.')          # from
practicum/geometry to practicum/kit/
import sympy as sp           # the escape
hatch: anything not in kit is in sp
from kit import *             # checks +
vec, plane, cross, intersection, unknown

language('en')               # this
notebook is in English, and so are the checks

# plane(Eq(2*x - 3*y - z, 6)), plane(A, n),
plane(A, B, C), plane(L1, L2), plane(L, A),
plane(R, P1, P2).
# cross(u, v) is the vector product;
P1.normal is the normal of the plane P1.
# A plane as an answer: Eq(2*x - 3*y - z, 6),
or vec(a) + lam*vec(b) + mu*vec(c).
# intersection(P1, P2) is what the planes
share: a point, a line, or None.

print('ready; sympy', sp.__version__)
A, B, C = vec(2, 1, 0), vec(0, 3, 1), vec(1,
0, 2)
print('AB  $\times$  AC:      ', cross(B - A, C -
A))
print('the plane ABC:      ', plane(A, B, C))
print('where it meets z = 0:',
intersection(plane(A, B, C), plane(Eq(z,
0))))

```

Map of the seven techniques

#	technique	you recognise it by	it reduces to
1	a point and a normal	<i>verify that A lies on the plane, P_2 is perpendicular to P_1, the plane meets the axes</i>	$\mathbf{r} \cdot \mathbf{n} = \mathbf{a} \cdot \mathbf{n}$
2	a normal from two directions	<i>the plane through A, B and C, contains L_1 and L_2, perpendicular to both planes</i>	$\mathbf{n} = \mathbf{u} \times \mathbf{v}$
3	a line and a plane	<i>the line meets the plane at P, verify that the line lies in the plane</i>	the general point of the line into the plane
4	two planes	<i>the line of intersection of Π_1 and Π_2</i>	direction $\mathbf{n}_1 \times \mathbf{n}_2$, one common point
5	three planes, a system	<i>the point where the three planes meet, no unique solution, intersect in a line</i>	eliminate, look at the last row

#	technique	you recognise it by	it reduces to
6	the perpendicular	<i>the foot of the perpendicular, the distance from L to Π, closest to the origin</i>	the line $\mathbf{b} + \lambda \mathbf{n}$ into the plane
7	the reflection	<i>the reflection of B in the plane, the reflected line</i>	twice the step to the foot

Techniques 1–2 build a plane. Techniques 3–5 ask what it shares with a line and with other planes — and the answer is always a set: a point, a line, or nothing. Techniques 6–7 walk from a point to the plane along its normal.

Part 1. A plane and its normal

Theory: a point and a normal

A plane is fixed by **one point** $\mathbf{a}$ on it and **one direction** $\mathbf{n}$ across it, the normal. A point $\mathbf{r}$ is on the plane when $\mathbf{r} - \mathbf{a}$ is perpendicular to $\mathbf{n}$:

$$\mathbf{r} \cdot \mathbf{n} = \mathbf{a} \cdot \mathbf{n}$$

Through $(1, 4, -2)$ with normal $(2, -1, 3)$: the right-hand side is the left-hand side worked out at the point, $2 - 4 - 6 = -8$, so

$$2x - y + 3z = -8$$

The normal is read off any Cartesian equation: the coefficients of x , y and z . Multiplying the whole equation by a number, $4x - 2y + 6z = -16$, gives the same plane.

A point is on the plane when it satisfies the equation: $(0, -1, -3)$ gives $0 + 1 - 9 = -8$, so it is; $(1, 1, 1)$ gives 4 , so it is not.

Where the plane meets an axis the other two coordinates are zero: the y -axis at $-y = -8$, the point $(0, 8, 0)$.

Two planes are parallel when their normals are multiples: $4x - 2y + 6z = 5$ is parallel to this one. **They are perpendicular** when their normals are: $x + 5y + z = 0$ has $2 - 5 + 3 = 0$. Do not swap the two tests. With a letter the test becomes an equation: $ax + 2y - z = 1$ is perpendicular to our plane when $2a - 2 - 3 = 0$, $a = \frac{5}{2}$.

A line normal to the plane uses the normal as its direction: $\mathbf{r} = (1, 4, -2) + t(2, -1, 3)$ — with " $\mathbf{r} =$ ", or the markscheme gives A0.

Task 1 — **May 2024 TZ2 Paper 1 Q11(a)-(c), 8 marks**

The plane P_1 has equation $2x + 6y - 2z = 5$.

(a) Verify that the point $A(2, \frac{1}{2}, 1)$ lies on the plane P_1 .

The plane P_2 is given by $(k^2 - 6)x + (2k + 3)y + pz = q$, where $p, q, k \in \mathbb{R}$ and $p \neq 0$.

(b) In the case where $p = -6$, P_2 is perpendicular to P_1 and A lies on P_2 . Find the value of k and the value of q .

For parts (c), (d) and (e) it is now given that P_2 is parallel to P_1 with $k = 3$.

(c) Determine the value of p .

Part (a) is one substitution — do it on paper. Parts (d) and (e) are in the archive.

```
k, p, q = symbols('k p q')

q1b = [...]      # [k, q]
q1c = ...        # p

A = vec(2, Rational(1, 2), 1)
P1 = plane(Eq(2*x + 6*y - 2*z, 5), name='P1')
P2 = plane(Eq((k**2 - 6)*x + (2*k + 3)*y +
p*z, q), name='P2')

verify_find('1b', q1b, [k, q],
[perpendicular(P1, P2.subs(p, -6)), on(A,
P2.subs(p, -6))])
verify_find('1c', q1c, p, [parallel(P1,
P2.subs(k, 3))])
```

Theory: a normal from two directions

Usually the question does not give the normal. It gives **two directions that lie in the plane**, and the normal is the direction perpendicular to both — their vector product:

$$\mathbf{u} \times \mathbf{v} = \begin{pmatrix} u_2v_3 - u_3v_2 \\ u_3v_1 - u_1v_3 \\ u_1v_2 - u_2v_1 \end{pmatrix}$$

Through three points $P(1, 0, 1)$, $Q(2, 2, 1)$, $R(1, 1, 4)$: the directions are $\overrightarrow{PQ} = (1, 2, 0)$ and $\overrightarrow{PR} = (0, 1, 3)$, and

$$\overrightarrow{PQ} \times \overrightarrow{PR} = \begin{pmatrix} 2 \cdot 3 - 0 \cdot 1 \\ 0 \cdot 0 - 1 \cdot 3 \\ 1 \cdot 1 - 2 \cdot 0 \end{pmatrix} = \begin{pmatrix} 6 \\ -3 \\ 1 \end{pmatrix}$$

Check it: $(6, -3, 1) \cdot (1, 2, 0) = 0$ and $(6, -3, 1) \cdot (0, 1, 3) = 0$. Then one point fixes the right-hand side, $6 - 0 + 1 = 7$, and the other two points must give the same 7: Q gives $12 - 6 + 1$. The plane is $6x - 3y + z = 7$.

The middle component is $u_3v_1 - u_1v_3$. Written the other way round it gives $(6, 3, 1)$, and Q then gives 19, not 7 — which is why the check with the other points is worth its ten seconds.

the question gives	the two directions
three points A, B, C	$\overrightarrow{AB}$ and $\overrightarrow{AC}$
a line and a point off it	the line's direction and the vector from a point of the line to the point
two lines that meet	their two directions
a plane perpendicular to two planes	the two normals

The vector form keeps the two directions instead of the normal:

$$\mathbf{r} = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} + \mu \begin{pmatrix} 0 \\ 1 \\ 3 \end{pmatrix}$$

Task 2 — **November 2021 Paper 2 Q11(a), 7 marks**

Three points $A(3, 0, 0)$, $B(0, -2, 0)$ and $C(1, 1, -7)$ lie on the plane Π_1 .

(a) (i) Find the vector $\overrightarrow{AB}$ and the vector $\overrightarrow{AC}$.

(ii) Hence find the equation of Π_1 , expressing your answer in the form $ax + by + cz = d$, where $a, b, c, d \in \mathbb{Z}$.

Enter a plane as `Eq(a*x + b*y + c*z, d)`.

```
q2a_AB = ...      # AB
q2a_AC = ...      # AC
q2a_ii = ...      # the plane: Eq(a*x + b*y +
c*z, d)

A, B, C = vec(3, 0, 0), vec(0, -2, 0), vec(1,
1, -7)

verify_find('2a(i) AB', q2a_AB, B - A)
verify_find('2a(i) AC', q2a_AC, C - A)
verify_plane('2a(ii)', q2a_ii, plane(A, B,
C))
```

Task 3 — **November 2025 TZ1** **Paper 2 Q12(b), (c), (e), 7 marks**

The equations of two lines, L_1 and L_2 , are given by:

$$L_1 : \mathbf{r}_1 = \begin{pmatrix} 5 \\ 4 \\ 2 \end{pmatrix} + s \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} \quad \text{where } s \in \mathbb{R}$$

$$L_2 : \frac{x+1}{2} = y-7 = \frac{z+5}{3}$$

The position vector of the point of intersection of L_1 and L_2 is $\begin{pmatrix} 1 \\ 8 \\ -2 \end{pmatrix}$ (part (a), technique 6 of C5).

The plane Π contains the lines L_1 and L_2 .

(b) (i) Write down the equation of Π , giving your answer in the form $\mathbf{r} = \mathbf{a} + \lambda\mathbf{b} + \mu\mathbf{c}$ where $\lambda, \mu \in \mathbb{R}$.

(ii) Given that $\mathbf{b} \times \mathbf{c} = \begin{pmatrix} -4 \\ -1 \\ 3 \end{pmatrix}$, show that the Cartesian equation of Π is $4x + y - 3z = 18$.

The plane intersects the coordinate axes at $P(4.5, 0, 0)$, $Q(0, q, 0)$ and $R(0, 0, r)$.

(c) Write down the value of **(i)** q ; **(ii)** r .

Another line, L_3 , is normal to Π and passes through the point of intersection of L_1 and L_2 .

(e) Write down an equation for L_3 in the form $\mathbf{r}_3 = \mathbf{m} + \gamma\mathbf{n}$.

For (b)(ii) enter $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})$, the right-hand side of $\mathbf{r} \cdot (\mathbf{b} \times \mathbf{c}) = \mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})$. In (e) use lam for γ . Parts (d), (f) and (g) are in C5 and C7.

```
q3b_i = ...      # vec(a) + lam*vec(b) +
mu*vec(c)
q3b_ii = ...     # a · (b × c)
q3c_i = ...      # q
q3c_ii = ...     # r
q3e = ...        # L3: vec(...) +
lam*vec(...)

s = symbols('s')
L1 = line(vec(5, 4, 2) + s*vec(1, -1, 1), s)
L2 = cartesian((x + 1)/2, y - 7, (z + 5)/3)
Pi = plane(L1, L2)
X = vec(1, 8, -2)                                #
where L1 and L2 meet
at_q, at_r = unknown('q'), unknown('r')

verify_plane('3b(i)', q3b_i, Pi)
verify_find('3b(ii)', q3b_ii, dot(X, vec(-4,
-1, 3)))
verify_find('3c(i)', q3c_i, at_q, [on(vec(0,
at_q, 0), Pi)])
verify_find('3c(ii)', q3c_ii, at_r,
[on(vec(0, 0, at_r), Pi)])
verify_line('3e', q3e, line(X, Pi.normal))
```

Task 4 — **May 2025 TZ1 Paper 1**
Q11(b), 4 marks

The plane P_1 has equation $x + 2y + z = 0$ and the plane P_2 has equation $x - y - 2z = 0$. (*Part (a), the acute angle between them, is in C7.*)

A third plane P_3 is perpendicular to both P_1 and P_2 .

The unique point of intersection of all three planes is the point $R(5, -5, 5)$.

(b) Find the Cartesian equation of P_3 .

Paper 1: by hand.

```
q4 = ...          # P3: Eq(...)

P1 = plane(Eq(x + 2*y + z, 0), name='P1')
P2 = plane(Eq(x - y - 2*z, 0), name='P2')
R = vec(5, -5, 5)

verify_plane('4', q4, plane(R, P1, P2))
```

Part 2. What lines and planes have in common

Theory: a line meets a plane

Put **the general point of the line** into the equation of the plane. For $\Pi: x + 2y - z = 4$ and $\mathbf{r} = (1, 0, 2) + \lambda(1, 1, 1)$ the general point is $(1 + \lambda, \lambda, 2 + \lambda)$:

$$(1 + \lambda) + 2\lambda - (2 + \lambda) = 4$$

$$\implies$$

$$2\lambda - 1 = 4$$

$$\implies$$

$$\lambda = \frac{5}{2}$$

and the point is the line at that value, $(\frac{7}{2}, \frac{5}{2}, \frac{9}{2})$. **The answer is the point**, not λ .

The equation in λ is linear, and it has three possible outcomes:

substituting gives	the line
one value of λ	meets the plane at one point
$4 = 4$, true for every λ	lies in the plane
$0 = 4$, true for no λ	is parallel to the plane and never meets it

$\mathbf{r} = (4, 0, 0) + \lambda(1, 0, 1)$ gives $(4 + \lambda) - \lambda = 4$ for every λ : it lies in Π . $\mathbf{r} = (0, 0, 0) + \lambda(1, 0, 1)$ gives $0 = 4$: parallel, no common point.

"Show that the line lies in the plane" needs the identity. One point of the line in the plane shows only that the line meets it.

Task 5 — **May 2025 TZ3 Paper 1 Q11(d), (f), 7 marks**

The points $A(1, -4, 0)$, $B(-3, -6, 2)$, $C(-1, -2, 4)$ and $D(3, 0, 2)$ form a parallelogram, $ABCD$. The diagonals $[AC]$ and $[BD]$ intersect at the point $E(0, -3, 2)$. It is given that $\overrightarrow{AB} \times \overrightarrow{AD} =$

$m \begin{pmatrix} -1 \\ 1 \\ -1 \end{pmatrix}$, where $m \in \mathbb{Z}^+$. (*D, E and m are parts (a)-(c): C5 and C7.*)

The plane, P_1 , contains the parallelogram $ABCD$.

(d) Find the Cartesian equation of P_1 .

A second plane, P_2 , has Cartesian equation $5x + y - 7z = 1$.

The line L passes through E and is perpendicular to P_1 .

The line L intersects the plane P_2 at point F .

(f) Find the coordinates of F .

For (f) enter the line L too.

```

q5d = ...          # P1: Eq(...)
q5f_L = ...        # L: vec(...) + lam*vec(...)
q5f = ...          # F

A, B, C, D, E = vec(1, -4, 0), vec(-3, -6,
2), vec(-1, -2, 4), vec(3, 0, 2), vec(0, -3,
2)
P1 = plane(A, B, D, name='P1')
P2 = plane(Eq(5*x + y - 7*z, 1), name='P2')
L = line(E, P1.normal)

verify_plane('5d', q5d, P1)
verify_line('5f L', q5f_L, L)
verify_intersection('5f', q5f, L, P2)

```

Theory: where two planes meet

Two planes that are not parallel meet in **a line**. It lies in both planes, so its direction is perpendicular to both normals:

$$\mathbf{d} = \mathbf{n}_1 \times \mathbf{n}_2$$

For $x + y + z = 6$ and $x - y + 2z = 5$:
 $(1, 1, 1) \times (1, -1, 2) = (3, -1, -2)$.

A point on both — set one coordinate to zero and solve the other two. With $y = 0$:
 $x + z = 6$ and $x + 2z = 5$ give $z = -1$, $x = 7$.
 So

$$\mathbf{r} = \begin{pmatrix} 7 \\ 0 \\ -1 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ -1 \\ -2 \end{pmatrix}$$

Or solve the two equations with one unknown left free. $z = t$: $x + y = 6 - t$ and $x - y = 5 - 2t$ give $x = \frac{11-3t}{2}$, $y = \frac{1+t}{2}$ — the line $(\frac{11}{2}, \frac{1}{2}, 0) + t(-\frac{3}{2}, \frac{1}{2}, 1)$. It is the same line: a different point, and a direction that is $-\frac{1}{2}$ of the one above.

"Verify that the line of intersection is ..." — substitute its general point into both equations. $x = 7 + 3\lambda$, $y = -\lambda$, $z = -1 - 2\lambda$ give $7 + 3\lambda - \lambda - 1 - 2\lambda = 6$ and $7 + 3\lambda + \lambda - 2 - 4\lambda = 5$ for every λ . One value of λ does not verify it: the markscheme gives (M1)A0.

The direction is not a normal. A normal sticks out of its plane; the line lies inside both.

Task 6 — **May 2023 TZ2 Paper 2 Q6(a), 3 marks**

Consider the two planes

$$\Pi_1 : 2x - y + 2z = 6 \quad \Pi_2 : 4x + 3y - z = 2$$

Let L be the line of intersection of Π_1 and Π_2 .

(a) Verify that a vector equation of L is

$$\mathbf{r} = \begin{pmatrix} 0 \\ 2 \\ 4 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ -2 \\ -2 \end{pmatrix}, \text{ where } \lambda \in \mathbb{R}.$$

Find both parts of the line from the planes, not from the line printed in the question: the point of L with $x = 0$, and the direction as $\mathbf{n}_1 \times \mathbf{n}_2$. Part (b) is in C7.

```
q6_point = ...          # the point of L where x
= 0
q6_direction = ...      # n1 x n2

P1 = plane(Eq(2*x - y + 2*z, 6), name='Π1')
P2 = plane(Eq(4*x + 3*y - z, 2), name='Π2')

verify_intersection('6 point', q6_point, P1,
P2, plane(Eq(x, 0)))
verify_direction('6 direction', q6_direction,
intersection(P1, P2))
```

Theory: three planes, and the system of equations they are

A system of three linear equations in x , y , z **is** three planes, and its solutions are the points on all three. So there are exactly three answers: one point, a whole line of points, or none.

$$x + y + z = 6$$

$$x - y + 2z = 5$$

$$2x + cz = e$$

Eliminate. The first two added give $2x + 3z = 11$. Take that from the third: $(c - 3)z = e - 11$. That is the last row, and the whole question is in it.

the last row	solutions	the planes
$c \neq 3$	one, $z = \frac{e-11}{c-3}$	meet at one point
$c = 3$ and $e = 11$: $0 = 0$	infinitely many	share a line
$c = 3$ and $e \neq 11$: $0 = e - 11$	none	have no common point

"No unique solution" asks when the coefficients of the last row vanish: $c = 3$.

"Infinitely many solutions" asks for

the right-hand side to vanish too: $e = 11$. Two different questions, and a question often asks both.

The general solution for $c = 3$, $e = 11$ is the line of the previous theory — one unknown becomes the parameter, the others are written through it.

A calculator solves a system with numbers, and the markscheme accepts *"attempt to solve using GDC"*. It cannot tell you when a row with a letter in it vanishes — that is done by hand.

A system with no solution looks several ways in space — all three planes parallel, two of them parallel, or three planes meeting in pairs along three parallel lines — and the algebra says only that there is no common point.

Task 7 — **May 2021 TZ1 Paper 2**
Q6, 5 marks

Consider the planes Π_1 and Π_2 with the following equations.

$$\Pi_1 : 3x + 2y + z = 6 \quad \Pi_2 : x - 2y + z = 4$$

(a) Find a Cartesian equation of the plane Π_3 which is perpendicular to Π_1 and Π_2 and passes through the origin $(0, 0, 0)$.

(b) Find the coordinates of the point where Π_1 , Π_2 and Π_3 intersect.

The check for (b) builds Π_3 itself, so a slip in (a) costs (a) only.

```
q7a = ...          # Π3: Eq(...)
q7b = ...          # the point, exact

P1 = plane(Eq(3*x + 2*y + z, 6), name='Π1')
P2 = plane(Eq(x - 2*y + z, 4), name='Π2')
P3 = plane(vec(0, 0, 0), P1, P2, name='Π3')

verify_plane('7a', q7a, P3)
verify_intersection('7b', q7b, P1, P2, P3)
```

Task 8 — May 2025 TZ2 Paper 2 Q9, 8 marks

A line L_1 has vector equation $\mathbf{r} = \begin{pmatrix} 0 \\ 0 \\ 2 \end{pmatrix} + t \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$ where $t \in \mathbb{R}$.

The plane P_1 contains the line L_1 and passes through the point $(2, 1, 5)$.

(a) Show that the Cartesian equation of the plane P_1 is $x + y - z = -2$.

Consider the three planes

$$P_1 : x + y - z = -2$$

$$P_2 : 2x + by - z = 3$$

$$P_3 : x - y + 2z = d$$

where $b, d \in \mathbb{R}^+$.

The three planes intersect in a line.

(b) Find the value of b and the value of d .

For (a) enter the normal you get from the line and the point, before you compare it with the printed equation.

```

b, d = symbols('b d')

q8a = ...          # a normal of P1
q8b = [...]        # [b, d]

L1 = line(vec(0, 0, 2) + t*vec(1, 0, 1), t)
P1 = plane(Eq(x + y - z, -2), name='P1')
P2 = plane(Eq(2*x + b*y - z, 3), name='P2')
P3 = plane(Eq(x - y + 2*z, d), name='P3')

verify_normal_vector('8a', q8a, plane(L1,
vec(2, 1, 5)))
verify_find('8b', q8b, [b, d],
[meet_in_line(P1, P2, P3), b > 0, d > 0])

```

Task 9 — November 2025 TZ3 Paper 2 Q7, 6 marks

Consider the following two equations.

$$2x + 6y - 8z = 13 \quad 3x - y + 3z = 12$$

(a) Determine the general solution giving the answer in parametric form.

Consider a third equation $ax + 12y - 16z = k$, where $a, k \in \mathbb{R}$.

(b) In the case where $a = 3$ and $k = -3$, determine the unique solution to the system of three equations.

(c) (i) Write down the value of a for which there is no unique solution to the

system of three equations.

(ii) Hence, write down the corresponding value of k for which there is an infinite number of solutions to the system of three equations.

In (a) enter the solution as `vec(x, y, z)` in terms of `t`. Each equation is a plane, and the checks treat it as one.

```
q9a = ...          # vec(x, y, z) in terms of t
q9b = ...          # (x, y, z)
q9c_i = ...        # a
q9c_ii = ...       # k

a, k = unknown('a'), unknown('k')
first = plane(Eq(2*x + 6*y - 8*z, 13))
second = plane(Eq(3*x - y + 3*z, 12))
third = plane(Eq(a*x + 12*y - 16*z, k))

verify_intersection('9a', q9a, first, second)
verify_intersection('9b', q9b, first, second,
third.subs({a: 3, k: -3}))
verify_find('9c(i)', q9c_i, a,
[no_unique_meet(first, second, third)])
verify_find('9c(ii)', q9c_ii, k,
[meet_in_line(first, second, third)])
```

Part 3. The perpendicular from a point

Theory: the foot of the perpendicular, and the distance

From $Q(4, 1, 5)$ to the plane $2x - y + 2z = -1$, walk **along the normal**:

$$\mathbf{r} = \begin{pmatrix} 4 \\ 1 \\ 5 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix}$$

and this line meets the plane where technique 3 says:

$$\begin{aligned} & 2(4 + 2\lambda) \\ & \quad - (1 - \lambda) \\ & \quad + 2(5 + 2\lambda) = \\ & \quad - 1 \\ & \implies \\ & 17 + 9\lambda = -1 \\ & \implies \\ & \lambda = -2 \end{aligned}$$

The foot is the line at $\lambda = -2$: $(0, 3, 1)$.

The distance is the length of the step,
 $|\lambda| |\mathbf{n}| = 2 \cdot 3 = 6$ — not $|\lambda| = 2$, which counts normals, not units.

The same number comes from
 $\frac{|\mathbf{n} \cdot \mathbf{q} - d|}{|\mathbf{n}|} = \frac{|17 + 1|}{3}$. Without the division it is 18, three times too long.

The point of a plane closest to the origin is the foot of the perpendicular from O : the line is $\lambda \mathbf{n}$.

A line parallel to a plane, or a parallel plane, is at the same distance from it everywhere: take any one of its points and drop the perpendicular from there.

Task 10 — **May 2022 TZ1 Paper 1 Q11, 15 marks**

Consider the three planes

$$\Pi_1 : 2x - y + z = 4$$

$$\Pi_2 : x - 2y + 3z = 5$$

$$\Pi_3 : -9x + 3y - 2z = 32$$

(a) Show that the three planes do not intersect.

(b) (i) Verify that the point $P(1, -2, 0)$ lies on both Π_1 and Π_2 .

(ii) Find a vector equation of L , the line of intersection of Π_1 and Π_2 .

(c) Find the distance between L and Π_3 .

For (a) enter what the three planes share — a point, a line or 'none' — and the value of $-9x + 3y - 2z$ along the line where Π_1 and Π_2 meet. Paper 1: (c) is exact.

```

q10a = ...          # a point, a line, or
'none'
q10a_value = ...    # -9x + 3y - 2z along
the line where Π1 and Π2 meet
q10b = ...          # L: vec(...) +
lam*vec(...)
q10c = ...          # the distance between L
and Π3, exact

P1 = plane(Eq(2*x - y + z, 4), name='Π1')
P2 = plane(Eq(x - 2*y + 3*z, 5), name='Π2')
P3 = plane(Eq(-9*x + 3*y - 2*z, 32),
name='Π3')
L = intersection(P1, P2)

verify_intersection('l0a', q10a, P1, P2, P3)
verify_find('l0a value', q10a_value,
P3.side(L.at(lam)))
verify_intersection('l0b(ii)', q10b, P1, P2)
verify_distance('l0c', q10c, L, P3,
exact=True)

```

Theory: the reflection in a plane

The foot of the perpendicular is **halfway** between a point and its reflection. From $Q(4,1,5)$ the foot was at $\lambda = -2$; the reflection is at **twice** that, $\lambda = -4$:

$$Q' = \begin{pmatrix} 4 \\ 1 \\ 5 \end{pmatrix} - 4 \begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix} = \begin{pmatrix} -4 \\ 5 \\ -3 \end{pmatrix}$$

Check it: the midpoint of QQ' is $(0, 3, 1)$, the foot. Or directly, $Q' = 2F - Q$.

A reflected line. The line $\mathbf{r} = (4, 1, 5) + t(1, 2, -3)$ meets the plane at $t = 3$, the point $M(7, 7, -4)$, and M **does not move** in the mirror. Any other point of the line — Q is one — goes to its reflection Q' . The reflected line runs through M and Q' :

$$\mathbf{r} = \begin{pmatrix} 7 \\ 7 \\ -4 \end{pmatrix} + \mu \begin{pmatrix} -11 \\ -2 \\ 1 \end{pmatrix}$$

Its direction is not the old one: keeping $(1, 2, -3)$ would give a line through M that stays on the same side.

A parallel plane on the other side.

The plane $2x - y + 2z = 17$ through Q is 6 units from $2x - y + 2z = -1$. The plane 6 units away on the other side keeps the normal and passes through Q' : $2(-4) - 5 + 2(-3) = -19$, so $2x - y + 2z = -19$. Changing the sign of 17 gives -17 , a plane $\frac{16}{3}$ units away.

Task 11 ● — **November 2023 TZ1**
Paper 2 Q12(b)-(d), 13 marks

Line L is given by the vector equation

$$\mathbf{r}_1 = \begin{pmatrix} 1 \\ 2 \\ -3 \end{pmatrix} + s \begin{pmatrix} 2 \\ 3 \\ 6 \end{pmatrix} \text{ where } s \in \mathbb{R}.$$

Line M is given by the vector equation

$$\mathbf{r}_2 = \begin{pmatrix} 9 \\ 9 \\ 11 \end{pmatrix} + t \begin{pmatrix} 4 \\ 1 \\ 2 \end{pmatrix} \text{ where } t \in \mathbb{R}.$$

(Part (a), where L and M meet, is the timed task of C5.)

(b) Verify that the lines L and M both lie in the plane P given by $\mathbf{r} \cdot \begin{pmatrix} 0 \\ 2 \\ -1 \end{pmatrix} = 7$.

Point B has position vector $\begin{pmatrix} -3 \\ 12 \\ 2 \end{pmatrix}$. A line through B perpendicular to P intersects P at point C .

(c) (i) Find the position vector of C .

(ii) Hence, find $|\overrightarrow{BC}|$.

(d) Find the reflection of the point B in the plane P .

For (b) enter $\mathbf{r} \cdot (0, 2, -1)$ at the general point of each line, in terms of its parameter: the check tells you whether it is 7 for every value.

```
q11b_L = ...      #  $\mathbf{r} \cdot (0, 2, -1)$  at the
general point of L, in terms of s
q11b_M = ...      # the same for M, in terms
of t
q11c_i = ...      # C
q11c_ii = ...     #  $|\mathbf{BC}|$ 
q11d = ...        # the reflection of B

s = symbols('s')
L = line(vec(1, 2, -3) + s*vec(2, 3, 6), s)
M = line(vec(9, 9, 11) + t*vec(4, 1, 2), t)
n = vec(0, 2, -1)
plane_P = plane(Eq(dot(vec(x, y, z), n), 7))
B = vec(-3, 12, 2)
C, image = unknown('C', 3), unknown('image',
3)

verify_find('11b L', q11b_L, dot(L.at(s), n))
verify_find('11b M', q11b_M, dot(M.at(t), n))
verify_find('11c(i)', q11c_i, C, [on(C,
plane_P), perpendicular(C - B, plane_P)])
verify_distance('11c(ii)', q11c_ii, B,
plane_P)
verify_find('11d', q11d, image,
[reflection(image, B, plane_P)])
```

Task 12 ● — November 2021 Paper 2
Q11(b)-(d), 14 marks

Three points $A(3, 0, 0)$, $B(0, -2, 0)$ and $C(1, 1, -7)$ lie on the plane Π_1 , and Π_1 has equation $2x - 3y - z = 6$ (part (a), Task 2).

Plane Π_2 has equation $3x - y + 2z = 2$.

(b) The line L is the intersection of Π_1 and Π_2 . Verify that the vector equation of

$$L \text{ can be written as } \mathbf{r} = \begin{pmatrix} 0 \\ -2 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}.$$

(c) The plane Π_3 is given by $2x - 2z = 3$. The line L and the plane Π_3 intersect at the point P .

(i) Show that at the point P , $\lambda = \frac{3}{4}$.

(ii) Hence find the coordinates of P .

(d) The point $B(0, -2, 0)$ lies on L .

(i) Find the reflection of the point B in the plane Π_3 .

(ii) Hence find the vector equation of the line formed when L is reflected in the plane Π_3 .

For (b) enter $\mathbf{n}_1 \times \mathbf{n}_2$. The check for (d)(ii) knows the line L and the plane Π_3 , not your reflection from (d)(i).

```
q12b = ...          # n1 x n2
q12c_i = ...        # λ where L meets Π3
q12c_ii = ...       # the point where L meets Π3
q12d_i = ...        # the reflection of B
q12d_ii = ...       # the reflected line:
vec(...) + mu*vec(...)

P1 = plane(Eq(2*x - 3*y - z, 6), name='Π1')
P2 = plane(Eq(3*x - y + 2*z, 2), name='Π2')
P3 = plane(Eq(2*x - 2*z, 3), name='Π3')
L = line(vec(0, -2, 0) + lam*vec(1, 1, -1))
B = vec(0, -2, 0)
at_P, image = unknown('lambda'),
unknown('image', 3)

verify_direction('12b', q12b,
intersection(P1, P2))
verify_find('12c(i)', q12c_i, at_P,
[on(L.at(at_P), P3)])
verify_intersection('12c(ii)', q12c_ii, L,
P3)
verify_find('12d(i)', q12d_i, image,
[reflection(image, B, P3)])
verify_line('12d(ii)', q12d_ii, mirror(L,
P3))
```

Trainer: name the technique in five seconds

Twelve openings. Do not compute anything — say only **which move you would make first**.

code	technique
plane	a point and a normal: an equation, a point on it, parallel or perpendicular planes
normal	a normal from two directions: the vector product
line_plane	the general point of a line into a plane
two_planes	the line where two planes meet
three_planes	three planes, a system of equations, a letter in it
foot	the perpendicular from a point: the foot, the distance
reflection	the reflection of a point or a line in a plane

1. Find the equation of the plane through $(1, -1, 2)$ with normal $(3, 0, -4)$.
2. Find the equation of the plane containing the points $(0, 1, 1)$, $(2, 0, 3)$

and $(1, 1, 0)$.

3. Find where the line $\mathbf{r} = (2, 1, 0) + \lambda(1, -1, 3)$ meets the plane $x + y + z = 9$.
4. Find a vector equation of the line where $x + 2y - z = 1$ and $2x - y + z = 7$ meet.
5. For which c does the system $x + y = 2$, $y + z = 3$, $x + cz = 1$ have no unique solution?
6. Find the distance from $(4, 0, -1)$ to the plane $2x + y - 2z = 3$.
7. Find the image of the point $(1, 2, 5)$ in the plane $z = 1$.
8. Show that the plane $3x - y + 2z = 4$ is perpendicular to the plane $x + 5y + z = 0$.
9. Find a normal to the plane containing $\mathbf{r} = \lambda(1, 1, 0)$ and $\mathbf{r} = \mu(0, 1, 1)$.
0. Show that the line $\mathbf{r} = (1, 1, 1) + \lambda(2, 0, -1)$ lies in the plane $x + 3y + 2z = 6$.
1. Determine whether the planes $x + y + z = 1$, $x - y = 0$ and $2x + z = 3$ meet at a single point.

2. Find the point of the plane $x - 2y + 2z = 18$ closest to the origin.

```
answers = {
    1: '', 2: '', 3: '', 4: '', 5: '', 6: '',
    7: '', 8: '', 9: '', 10: '', 11: '', 12:
    '',
}

trigger_check(answers, {1: 'fad900280ce5', 2:
'317b32c14369', 3: '01fe0f8f75b2', 4:
'cab053b1e3e1', 5: 'c2b6103cd165', 6:
'22e4f6912a69', 7: 'f30409c51a78', 8:
'fad900280ce5', 9: '317b32c14369', 10:
'01fe0f8f75b2', 11: 'c2b6103cd165', 12:
'22e4f6912a69'})
```

On the clock — *November 2022* *Paper 2 Q12(d), 9 marks*

Nine marks, thirteen minutes.
Calculator allowed, no hints.

Consider the points $A(1, 2, 3)$, $B(k, -2, 1)$ and $C(5, 0, 2)$, where $k \in \mathbb{R}$.

(d) For $k \neq 9$, let Π be the plane containing A , B and C .

(i) Find the Cartesian equation of the plane Π .

(ii) Find the coordinates of the point on the plane Π which is closest to the origin $(0, 0, 0)$.

Attempt log

date	time	result

```
qt_i = ...      #  $\Pi$ : Eq(...)
qt_ii = ...     # the point closest to the
origin

k = symbols('k')
Pi = plane(vec(1, 2, 3), vec(k, -2, 1),
vec(5, 0, 2))
closest = unknown('X', 3)

verify_plane('timer (i)', qt_i, Pi)
verify_find('timer (ii)', qt_ii, closest,
[on(closest, Pi), perpendicular(closest,
Pi)])
```



Solutions

Work these only after you have your own answer, or you are reading, not practising.

1 (a) $2 \cdot 2 + 6 \cdot \frac{1}{2} - 2 \cdot 1 = 5.$

1 (b) The normals are $(2, 6, -2)$ and $(k^2 - 6, 2k + 3, -6)$, and perpendicular planes have perpendicular normals:

$$\begin{aligned} 2(k^2 - 6) &+ 6(2k + 3) \\ &+ 12 &= 2k^2 \\ &+ 12k \\ &+ 18 = 2(k + 3)^2 = 0 \end{aligned}$$

$\implies$

$$\boxed{k = -3}$$

— one value, not two. Then P_2 is $3x - 3y - 6z = q$, and A on it gives $6 - \frac{3}{2} - 6 =$

$$\boxed{q = -\frac{3}{2}}.$$

1 (c) With $k = 3$ the normal is $(3, 9, p) = \frac{3}{2}(2, 6, -2)$ when $\boxed{p = -3}$.

$$2 \text{ (a)(i)} \quad \overrightarrow{AB} = \begin{pmatrix} -3 \\ -2 \\ 0 \end{pmatrix}, \quad \overrightarrow{AC} = \begin{pmatrix} -2 \\ 1 \\ -7 \end{pmatrix}$$

$$2 \text{ (a)(ii)} \quad \overrightarrow{AB} \times \overrightarrow{AC} = \begin{pmatrix} (-2)(-7) - 0 \cdot 1 \\ 0 \cdot (-2) - (-3)(-7) \\ (-3) \cdot 1 - (-2)(-2) \end{pmatrix} = \begin{pmatrix} 14 \\ -21 \\ -7 \end{pmatrix} = 7 \begin{pmatrix} 2 \\ -3 \\ -1 \end{pmatrix},$$

and A gives 6: $\boxed{2x - 3y - z = 6}$. Check with C : $2 + (-3) + 7 = 6$.

3 (b)(i) L_2 is $(-1, 7, -5) + \mu(2, 1, 3)$. A point of L_1 and the directions of both lines:

$$\mathbf{r} = \begin{pmatrix} 5 \\ 4 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} + \mu \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix}$$

3 (b)(ii) $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = (1, 8, -2) \cdot (-4, -1, 3) = -4 - 8 - 6 = \boxed{-18}$, so $-4x - y + 3z = -18$, which is $4x + y - 3z = 18$. Any point of Π gives the same -18 .

3 (c) $x = z = 0$: $y = \boxed{18}$; $x = y = 0$: $-3z = 18$, $\boxed{z = -6}$.

$$\mathbf{3 (e)} \quad \mathbf{r}_3 = \begin{pmatrix} 1 \\ 8 \\ -2 \end{pmatrix} + \gamma \begin{pmatrix} -4 \\ -1 \\ 3 \end{pmatrix}$$

$$\mathbf{4} \quad \mathbf{n}_1 \times \mathbf{n}_2 = (1, 2, 1) \times (1, -1, -2) = (-4 + 1, 1 + 2, -1 - 2) = (-3, 3, -3) = -3(1, -1, 1).$$

$$R \text{ gives } 5 + 5 + 5 = 15: \boxed{x - y + z = 15}.$$

$$\mathbf{5 (d)} \quad m(-1, 1, -1) \text{ is a normal, and } A \text{ gives } -1 - 4 - 0 = -5: \boxed{-x + y - z = -5}.$$

$$\mathbf{5 (f)} \quad L: \mathbf{r} = \begin{pmatrix} 0 \\ -3 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ 1 \\ -1 \end{pmatrix}. \text{ Into } P_2:$$

$$5(-\lambda) + (-3 + \lambda) - 7(2 - \lambda) = 3\lambda - 17 = 1, \\ \lambda = 6, \text{ and } F = \boxed{(-6, 3, -4)}.$$

6 With $x = 0$: $-y + 2z = 6$ and $3y - z = 2$ give $y = 2$, $z = 4$ — the point $\boxed{(0, 2, 4)}$, and it is on both planes. $\mathbf{n}_1 \times \mathbf{n}_2 = (2, -1, 2) \times (4, 3, -1) = \boxed{(-5, 10, 10)} = -5(1, -2, -2)$, perpendicular to both normals. So L is the printed line.

7 (a) $(3, 2, 1) \times (1, -2, 1) = (4, -2, -8) = 2(2, -1, -4)$, through the origin:
 $\boxed{2x - y - 4z = 0}$.

7 (b) Solving the three equations,
 $\boxed{\left(\frac{41}{21}, -\frac{10}{21}, \frac{23}{21}\right)} = (1.95, -0.476, 1.10)$.

8 (a) Two directions in P_1 : $(1, 0, 1)$ along L_1 and $(2, 1, 5) - (0, 0, 2) = (2, 1, 3)$. $(1, 0, 1) \times (2, 1, 3) = \boxed{(-1, -1, 1)}$, and $(0, 0, 2)$ gives -2 :
 $-x - y + z = 2$, which is $x + y - z = -2$.

8 (b) $P_2 - 2P_1$: $(b - 2)y + z = 7$. $P_3 - P_1$:
 $-2y + 3z = d + 2$. Three times the first
minus the second: $(3b - 4)y = 19 - d$. A line
of solutions needs $0 = 0$:

$$\boxed{b = \frac{4}{3}, \quad d = 19}$$

9 (a) Let $z = t$. Then $x = \frac{17}{4} - \frac{t}{2}$, $y = \frac{3}{4} + \frac{3t}{2}$:

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 17/4 - t/2 \\ 3/4 + 3t/2 \\ t \end{pmatrix}$$

9 (b) Into $3x + 12y - 16z = -3$: $\frac{87}{4} + \frac{t}{2} = -3$,
 $t = -\frac{99}{2}$, so $\boxed{(29, -73.5, -49.5)}$.

9 (c) $12y - 16z = 2(6y - 8z)$, so the third equation is twice the first when $\boxed{a = 4}$; then the right-hand side must be $2 \cdot 13$:
 $\boxed{k = 26}$.

10 (a) $\Pi_2 - 2\Pi_1$: $-3x + z = -3$. $\Pi_3 + 3\Pi_1$:
 $-3x + z = 44$. The same left side, two different right sides: no common point,
 $\boxed{\text{none}}$. Along the line where Π_1 and Π_2 meet, $-9x + 3y - 2z = \boxed{-15}$ for every λ , never 32.

10 (b) (i) $2 + 2 + 0 = 4$ and $1 + 4 + 0 = 5$.
 (ii) $(2, -1, 1) \times (1, -2, 3) = (-1, -5, -3)$:

$$\mathbf{r} = \begin{pmatrix} 1 \\ -2 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 5 \\ 3 \end{pmatrix}$$

10 (c) $(1, 5, 3) \cdot (-9, 3, -2) = 0$: L is parallel to Π_3 . From P along the normal, $(1 - 9t, -2 + 3t, -2t)$: $-15 + 94t = 32$, $t = \frac{1}{2}$, and the distance is $\frac{1}{2}\sqrt{94} = \boxed{\frac{\sqrt{94}}{2}}$.

11 (b) $2(2 + 3s) - (-3 + 6s) = \boxed{7}$ and $2(9 + t) - (11 + 2t) = \boxed{7}$, for every s and every t .

11 (c) $(-3, 12 + 2\lambda, 2 - \lambda)$ into P : $2(12 + 2\lambda) - (2 - \lambda) = 22 + 5\lambda = 7$, $\lambda = -3$: $\overrightarrow{OC} = \boxed{\begin{pmatrix} -3 \\ 6 \\ 5 \end{pmatrix}}$, and $|\overrightarrow{BC}| = 3|(0, 2, -1)| = \boxed{3\sqrt{5}} = 6.71$.

11 (d) $2\lambda = -6$: $\boxed{(-3, 0, 8)}$.

12 (b) $(2, -3, -1) \times (3, -1, 2) = \boxed{(-7, -7, 7)} = -7(1, 1, -1)$, and $(0, -2, 0)$ is on both: $6 = 6$, $2 = 2$.

12 (c) $2\lambda - 2(-\lambda) = 4\lambda = 3$, $\lambda = \frac{3}{4}$, and

$$P = \left(\frac{3}{4}, -\frac{5}{4}, -\frac{3}{4} \right).$$

12 (d)(i) From B along $(2, 0, -2)$: $(2\mu, -2, -2\mu)$ into Π_3 gives $8\mu = 3$, $\mu = \frac{3}{8}$ — the foot $(\frac{3}{4}, -2, -\frac{3}{4})$. Twice as far:

$$\left(\frac{3}{2}, -2, -\frac{3}{2} \right).$$

12 (d)(ii) P stays, B goes to B' : the direction $B' - P = (\frac{3}{4}, -\frac{3}{4}, -\frac{3}{4})$ and

$$\mathbf{r} = \begin{pmatrix} 3/2 \\ -2 \\ -3/2 \end{pmatrix} + \mu \begin{pmatrix} 3/4 \\ -3/4 \\ -3/4 \end{pmatrix}$$

Timer

(i) $\overrightarrow{AB} = (k-1, -4, -2)$, $\overrightarrow{AC} = (4, -2, -1)$ and $\overrightarrow{AB} \times \overrightarrow{AC} = (0, k-9, 18-2k) = (k-9)(0, 1, -2)$. Since $k \neq 9$ the normal is $(0, 1, -2)$, and A gives $2 - 6$: $y - 2z = -4$.

(ii) From O along the normal, $(0, \lambda, -2\lambda)$: $\lambda + 4\lambda = -4$, $\lambda = -\frac{4}{5}$, and the point is

$$\boxed{\left(0, -\frac{4}{5}, \frac{8}{5}\right)} = (0, -0.8, 1.6).$$